

Snowflake Virtual Warehouse Scaling, Caching, and Monitoring Assignment

Objective

The purpose of this assignment is to understand and demonstrate the following Snowflake features:

- Creating and configuring a Virtual Warehouse
 - Scaling warehouse resources and analyzing performance differences
 - Utilizing Snowflake caching mechanisms
 - Monitoring warehouse utilization and query performance metrics
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Part 1: Creating a Virtual Warehouse

Steps Performed -

1. Log in to Snowflake Web UI

Access the Snowflake Web Interface using valid credentials.

2. Navigate to Warehouses

From the left navigation panel, select the **Warehouses** tab.

3. Create a New Warehouse

Click **Create** and provide the following configuration:

Parameter	Value
Warehouse Name	TEST_WAREHOUSE
Size	Small
Auto-Suspend	5 Minutes
Auto-Resume	Enabled

4. Complete Creation

Click **Finish** to create the warehouse.

Expected Outcome

A new warehouse named **TEST_WAREHOUSE** is successfully created and ready to execute queries.

New warehouse

Owner: ACCOUNTADMIN

Temp_Warehouse

Add description

Type: **Standard (Gen2)** ▾ Size: **Small** ▾

Advanced options ^

☒ **Auto-resume**
Automatically resumes the warehouse when any statement that requires a warehouse is submitted.

☒ **Auto-suspend**
Automatically suspends the warehouse if it is inactive for the specified period of time.

Suspend after
 min(s) of inactivity

☐ **Multi-cluster warehouse**
Scale compute resources as query concurrency needs change.

☒ **Query acceleration**
Accelerate outlier queries with additional flexible compute resources.

Scale factor x2

[Learn more about pricing](#) Cancel Create

Part 2: Experimenting with Warehouse Scaling

Objective

Evaluate the impact of warehouse size on query execution performance.

Query Used

```
SELECT COUNT(*)  
FROM SNOWFLAKE_SAMPLE_DATA.TPCH_SF1.LINEITEM;
```

Procedure

Step 1: Execute Query Using Small Warehouse

Run the query using the newly created Small warehouse.

Step 2: Scale the Warehouse

1. Navigate to the **Warehouses** tab.
2. Locate **TEST_WAREHOUSE**.
3. Click **Edit**.
4. Change the warehouse size from **Small** to **Medium** or **Large**.
5. Save the changes.

Step 3: Execute the Same Query Again

Run the identical query after scaling.

Observations

Warehouse Size	Query Execution Time
Small	Record observed time
Medium/Large	Record observed time

Analysis

- Larger warehouses provide more compute resources.
- Query execution time generally decreases as warehouse size increases.
- Performance improvements are more noticeable for large datasets and complex workloads.

